

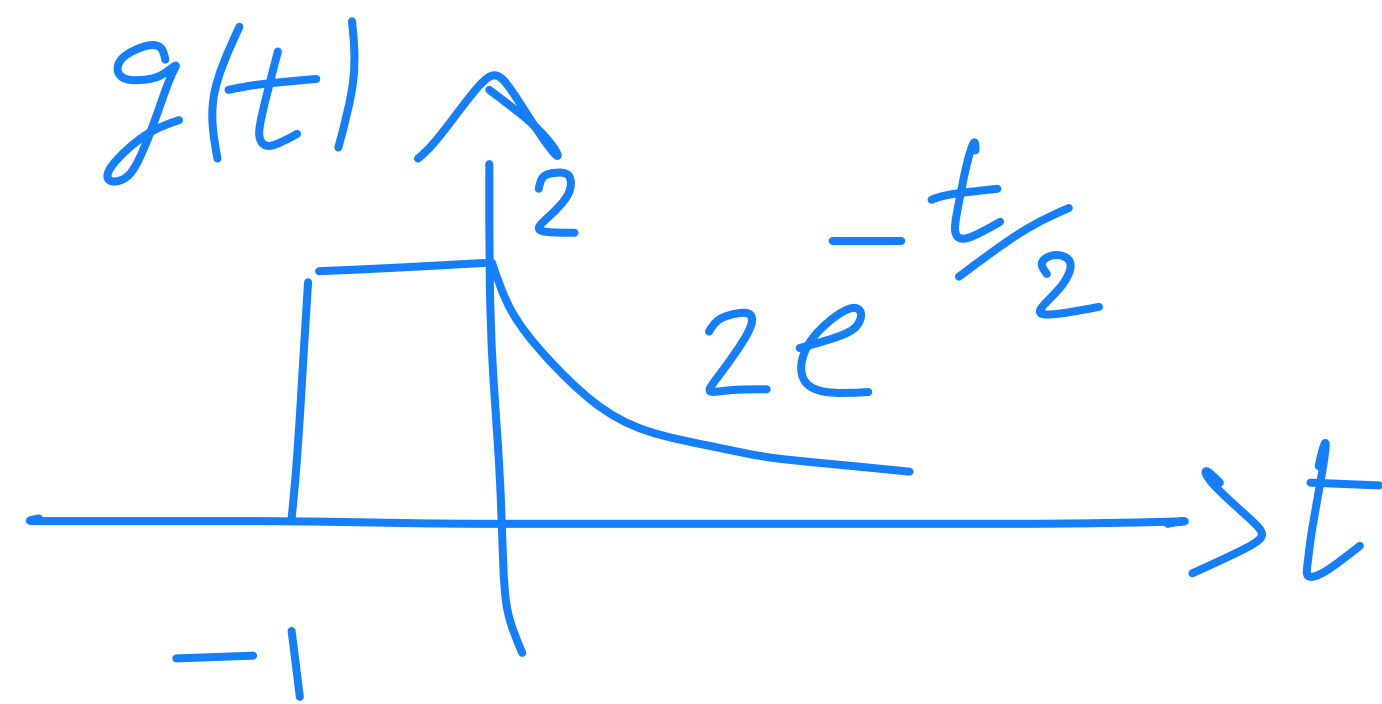
Principles of communication systems

EET3202, CUNY City Tech, Fall 2023

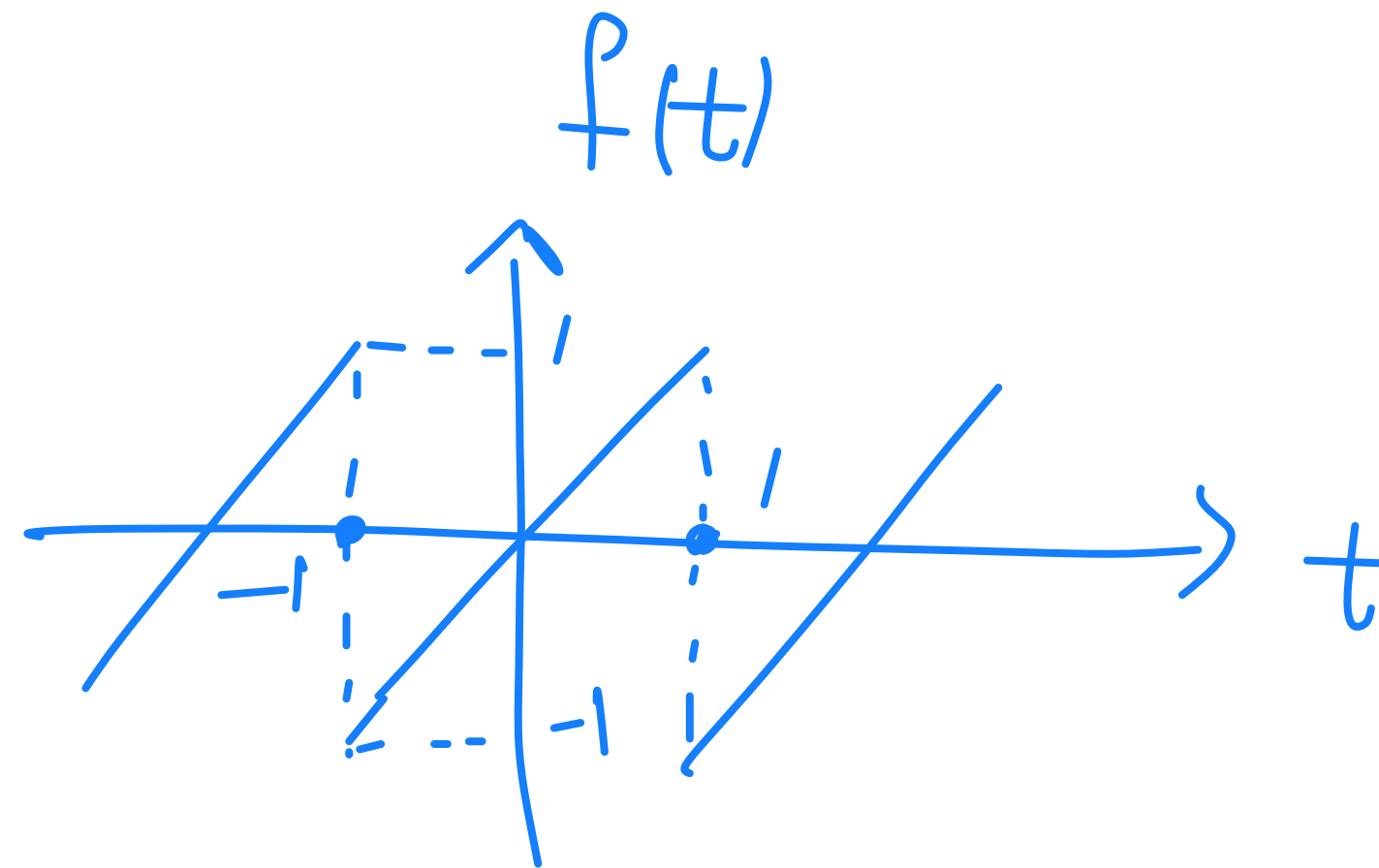
Homework #01 Solutions

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Problem 1



$$\begin{aligned}
 E_g &= \int_{-\infty}^{\infty} |g(t)|^2 dt = \int_{-1}^0 2^2 dt + \int_0^{\infty} 4e^{-t} dt \\
 &= 4(0 - (-1)) - 4e^{-t} \Big|_0^{\infty} \\
 &= 4 + 4 = 8
 \end{aligned}$$



$$\begin{aligned}
 P_f &= \frac{1}{2} \int_{-1}^1 |f(t)|^2 dt = \frac{1}{2} \int_{-1}^1 t^2 dt = \frac{1}{2} \left[\frac{1}{3} t^3 \Big|_{-1}^1 \right] \\
 &= \frac{1}{3}
 \end{aligned}$$

* $P_g = 0$, $E_f = \infty$

Problem 2

Our goal is to prove $\delta(at) = \frac{1}{|a|} \delta(t)$

we start by evaluating $\int_{-\infty}^{\infty} f(t) \delta(at) dt$ $\xrightarrow[\substack{\text{change of} \\ \text{variables} \\ x=at}]{}$ $\begin{cases} \int_{-\infty}^{+\infty} f(\frac{x}{a}) \delta(x) \cdot \frac{1}{a} dx, & a > 0 \\ \int_{+\infty}^{-\infty} f(\frac{x}{a}) \delta(x) \cdot \frac{1}{a} \cdot dx, & a < 0 \end{cases}$

note that the integral bounds are not exactly the same for $a > 0$ and $a < 0$, but we can combine the two cases:

$$\frac{1}{|a|} \int_{-\infty}^{\infty} f(\frac{x}{a}) \delta(x) dx = \frac{1}{|a|} f(0)$$

Problem 2

Continued...

So what we found out is that when we multiply a function $f(t)$ with $\delta(at)$ and integrate it, we obtain $\frac{1}{|a|}f(0)$ which is equivalent to

$$\int_{-\infty}^{\infty} f(t) \left[\frac{1}{|a|} \delta(t) \right] dt$$

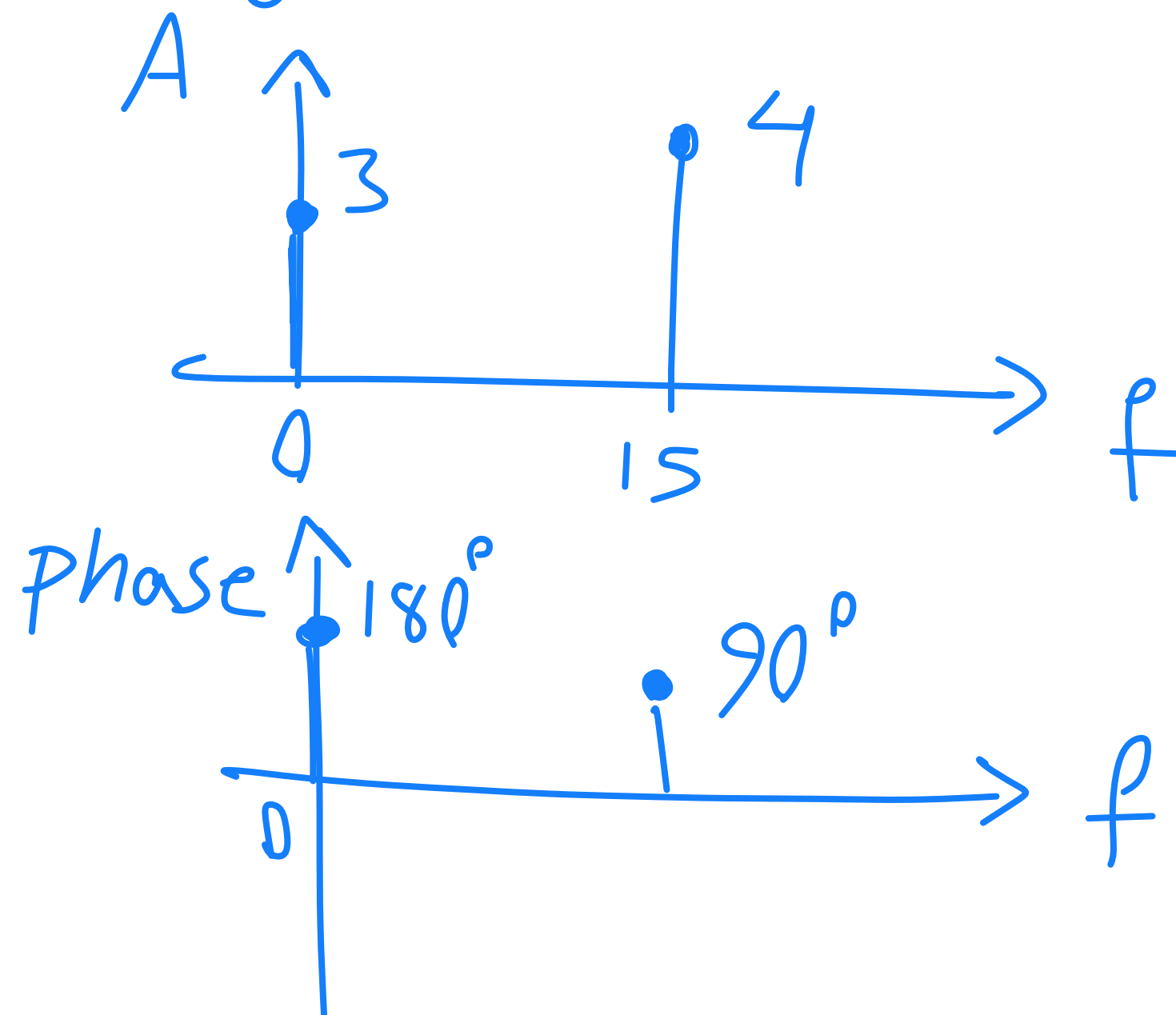
Therefore, we conclude that $\delta(at) = \frac{1}{|a|} \delta(t)$

Problem 3

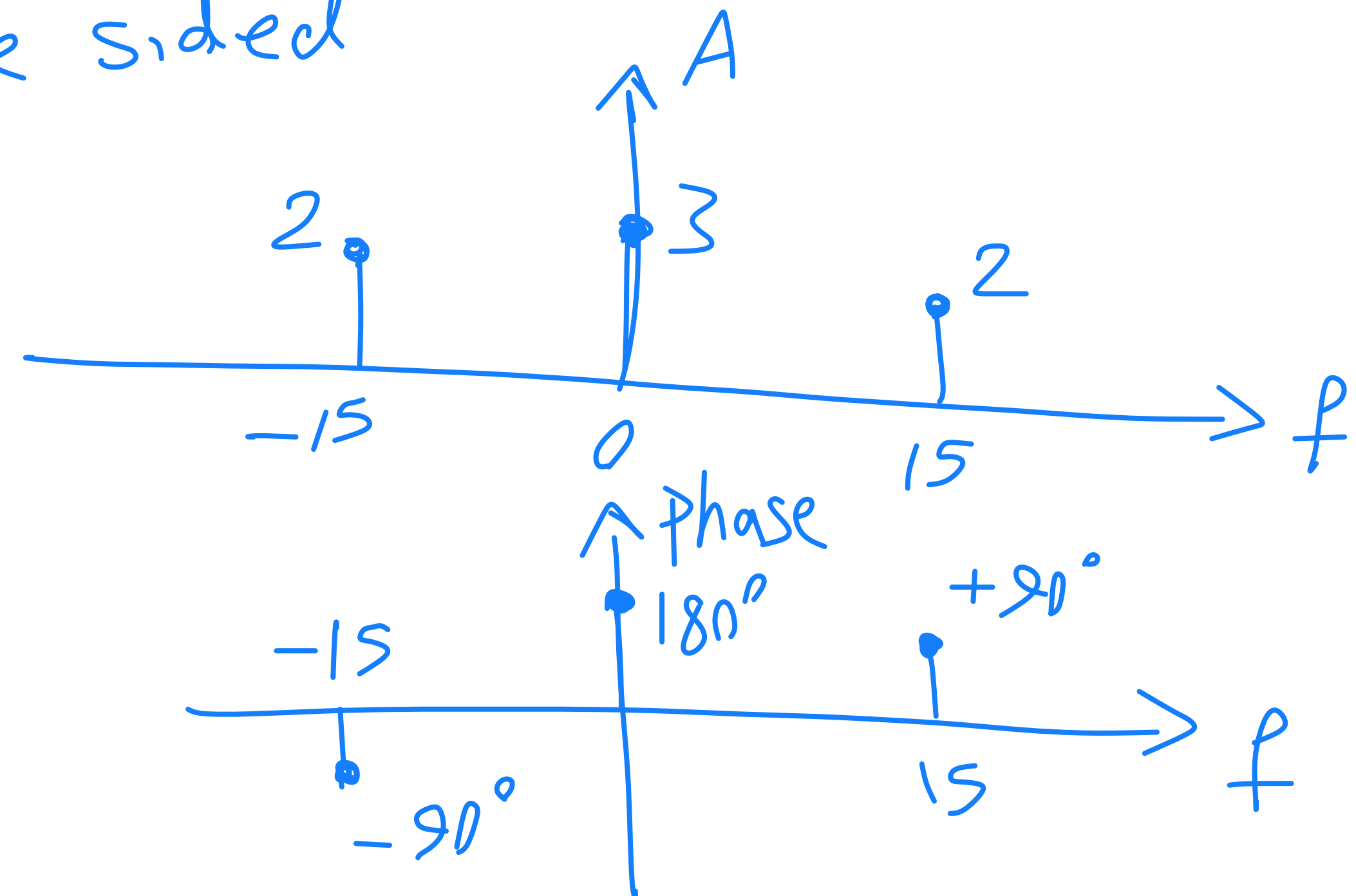
$$v(t) = -3 - 4 \sin(30\pi t)$$

$$= 3 \cos(2\pi(0)t + 180^\circ) + 4 \cos(2\pi(15)t - 90^\circ + 180^\circ)$$

Single sided



Double sided



Problem 4

$$v_n(t) = e^{j2\pi n f_0 t}, \quad v_m(t) = e^{j2\pi m f_0 t}$$

$$\langle v_n(t), v_m(t) \rangle = \int_0^{T_0} v_n(t) v_m^*(t) dt$$

↑ This should be conjugated.
I forgot the conjugation sign (*)
in the original problem set

$$\int_0^{T_0} e^{j2\pi(n-m)f_0 t} dt = \begin{cases} T_0 & , n=m \\ 0 & , n \neq m \end{cases}$$

Problem 5

$v(t)$ is real which means $v^*(t) = v(t)$

$$v(t) = \sum_{n=-\infty}^{\infty} C_n e^{j2\pi n f_0 t} \xrightarrow[\text{both sides}]{\text{conjugate}} v^*(t) = \sum_{n=-\infty}^{\infty} C_n^* e^{-j2\pi n f_0 t}$$

change of variable from n to $-n \rightarrow v^*(t) = \sum_{n=-\infty}^{\infty} C_{-n}^* e^{j2\pi n f_0 t}$

$$v^*(t) = v(t) \xrightarrow{\text{Therefore}} C_n = C_{-n}^* \quad \text{or} \quad C_n^* = C_{-n}$$

Problem 5

Continued...

$$C_n = |C_n| e^{j(\arg C_n)} = |C_n| e^{j\theta_n}$$

Let's rewrite $v(t)$:

$$v(t) = C_0 + C_1 e^{j2\pi(1)f_0 t} + C_{-1} e^{j2\pi(-1)f_0 t} + \dots$$

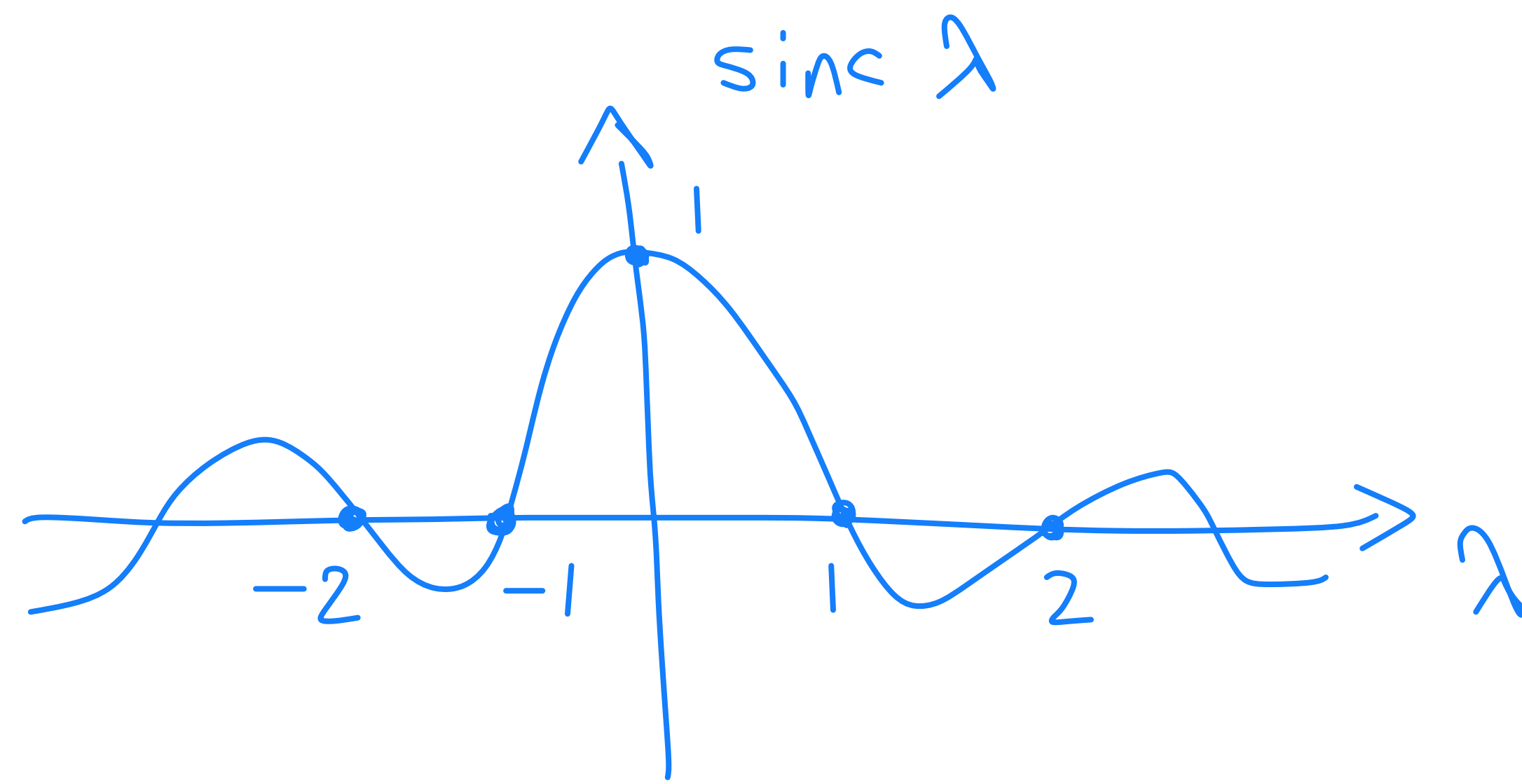
↑ from previous slide $C_{-1} = C_1^*$

$$= C_0 + C_1 e^{j2\pi(1)f_0 t} + C_1^* e^{-j2\pi f_0 t} + C_2 e^{j2\pi(2)f_0 t} + C_2^* e^{-j2\pi(2)f_0 t} + \dots$$

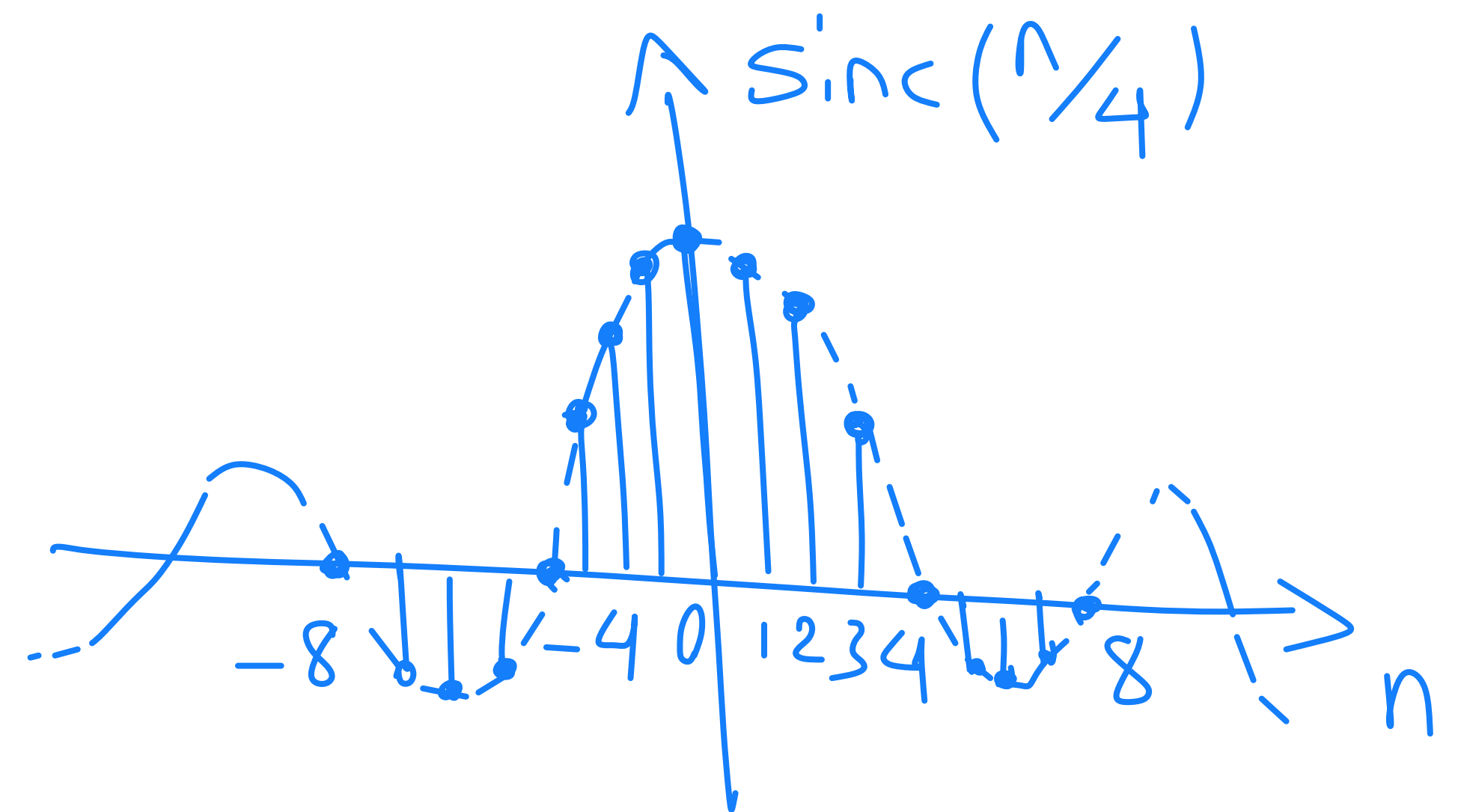
$$= C_0 + |C_1| \left(e^{j(2\pi(1)f_0 t + \theta_1)} + e^{-j(2\pi(1)f_0 t + \theta_1)} \right) + \dots$$

$$= C_0 + |C_1| 2 \cos(2\pi(1)f_0 t + \theta_1) + \dots = C_0 + \sum_{n=1}^{\infty} 2|C_n| \cos(2\pi n f_0 t + \theta_n)$$

Problem 6



Continuous-time function



Discrete-time function